

Indoor Greenhouse Robot Experiments

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1 Project Background and Integration

This project is based on the [RoverRobotics Repo](#), which provides all necessary infrastructure such as robot control, simulation in Gazebo, visualization in RViz, hardware interface, navigation, and SLAM.

The current work is connected to my own [Project's Repo](#).

1.1 Problems Addressed

- **Fixed SLAM Mapping Malfunction:**

Issue: The map rotated with the rover, making it unreadable.

Cause: The EKF relied only on missing camera/IMU data for yaw velocity.

Fix: Added yaw-velocity calculation from odometry.

- **Fixed Missing Rotation Recognition in Simulation:**

Same root cause and fix as the SLAM issue — yaw velocity is now estimated from odometry.

- **Fixed Wheel Lock on Boot:**

Issue: Wheels locked immediately after boot due to a startup service suggested by the manufacturer.

Fix: Removed the conflicting service.

Future: Investigate the service's purpose and re-integrate properly if needed.

Additionally, I added an IMU BNO055 component for improved odometry and localization results and utilized an Extended Kalman Filter (EKF) with the encoder sensor.

1.2 Project Overview

The goal of this project is to drive a robot in greenhouses to monitor plant diseases. Experiments are currently being conducted both indoors and outdoors using three rows of fake plants to simulate greenhouse conditions, with five plants per row.

1.2.1 Environment and Equipment

System: Ubuntu 22.04, ROS 2 Humble.

Sensors: Lidar S2 and IMU BNO055.

The indoor environment was scanned and named "indoor_abc_lab3". Each plant inception is considered as a 3-second stop next to the plant, either to the left or right of the robot.

1.3 Experiment Goals

Indoor Experiments:

- **Experiment 1:** Stop next to each plant.
- **Experiment 2:** Stop next to each second plant.
- **Experiment 3:** Stop next to each n -th plant (n is a user-supplied argument, different each time).
- **Experiment 4:** Stop next to five predetermined plants considered infected.
- **Experiment 5:** Stop next to each post-determined (after the robot starts "checking" plants, live push data of an infected plant in real-time) infected plant from m infected plants, and for each i from m infected plants, check the $j + 1$ and $j - 1$ plants (the plants before and after the infected plant row) for disease spread.

There should be three levels of infection: 10%, 25%, and 50%.

Each experiment is to be conducted three times, both with and without the IMU.

Thus, 5 experiments $\times$ 3 repeats $\times$ 2 (with/without IMU) equals 30 total experiments.

Outdoor Experiments:

The indoor process is to be repeated outdoors, resulting in another 30 experiments.

Total: Indoor + Outdoor = 60 experiments in total.

1.4 Current Status

At present, only indoor experiments with the IMU have been completed, without infection levels, in a simulation based on the real lab lidar scan. A Gazebo world was created based on the lidar map as well.

Experiments 1–4 were conducted, with three attempts for each experiment, totaling 12 experiments. Due to the war situation, the experiments were carried out only in simulation.

2 Overview

I ran four different "snake" pattern navigation experiments in a Gazebo indoor greenhouse. In each run the robot visits plant centers in the order 1→5→10→6→11→15, but stops at different subsets depending on the experiment type:

Experiment	Type	Description
1		Stop at <i>every</i> plant (IDs 1,2,3,4,5,10,9,8,7,6,11,12,13,14,15)
2		Stop at <i>every 2nd</i> plant
3		Stop at <i>every 3rd</i> plant
4		Stop at <i>predetermined</i> plants (user-supplied)

Each experiment was repeated **three** times to gauge repeatability.

3 Commands & Launch Files

3.1 Gazebo Simulation

```
ros2 launch roverrobotics_gazebo 2wd_rover_indoor_lab.launch.py
```

3.2 Navigation Stack

```
ros2 launch roverrobotics_driver navigation_launch.py
```

3.3 Run a Specific Experiment

```
ros2 run roverrobotics_driver greenhouse_experiment_node --ros-args  
-p experiment_type=<1|2|3|4>
```

3.4 Post-process Results

```
ros2 run experiment_monitor experiment_monitor --ros-args  
-p experiment_type=<1|2|3|4>
```

Replace <1|2|3|4> with the desired experiment type.

4 Key Results

Below are the aggregate statistics (mean $\pm$ std over three attempts) for each experiment type:

Exp	Mean task dur [s]	Std dur [s]	Mean error [m]	Std err [m]	Success %
1 (all)	13.3	3.7	0.19	0.004	100 %
2 (every second)	23.4	7.0	0.19	0.041	98 %
3 (every third)	24.7	13.1	0.18	0.011	100 %
4 (predetermined)	32.1	7.9	0.18	0.014	100 %

Table 1: Mean $\pm$ std across three attempts per experiment.

5 Experiment Visualizations

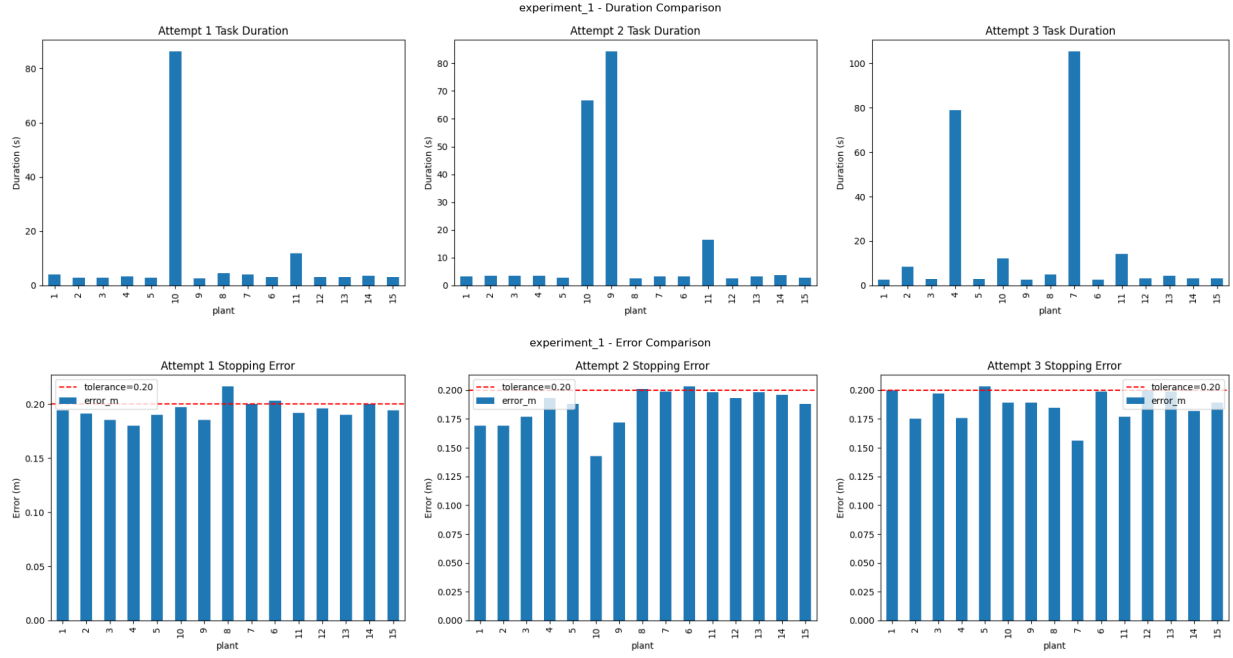


Figure 1: Experiment Type 1: Stop at every plant.

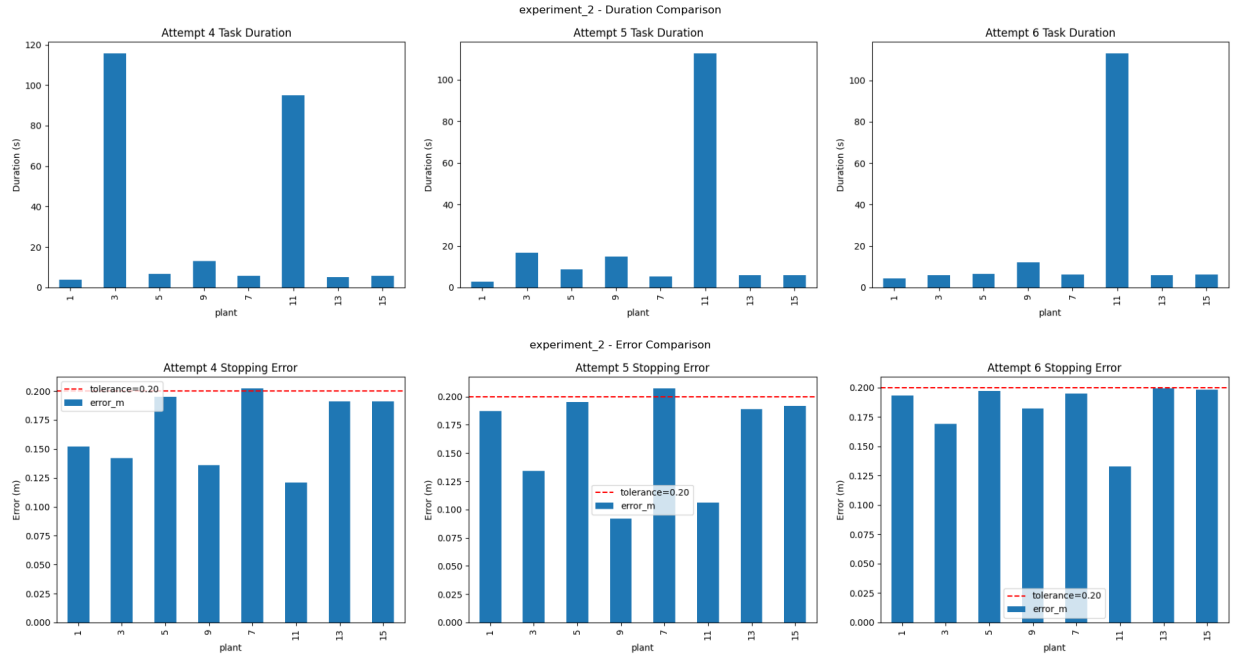


Figure 2: Experiment Type 2: Stop at every 2nd plant.

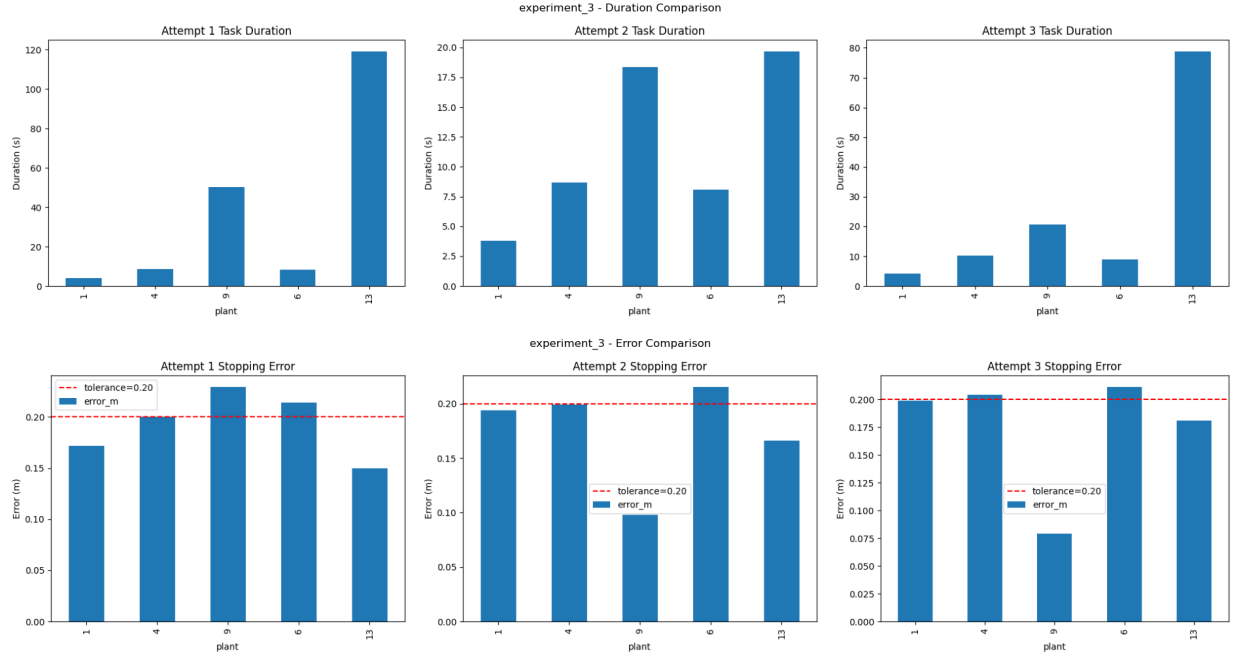


Figure 3: Experiment Type 3: Stop at every 3rd plant.

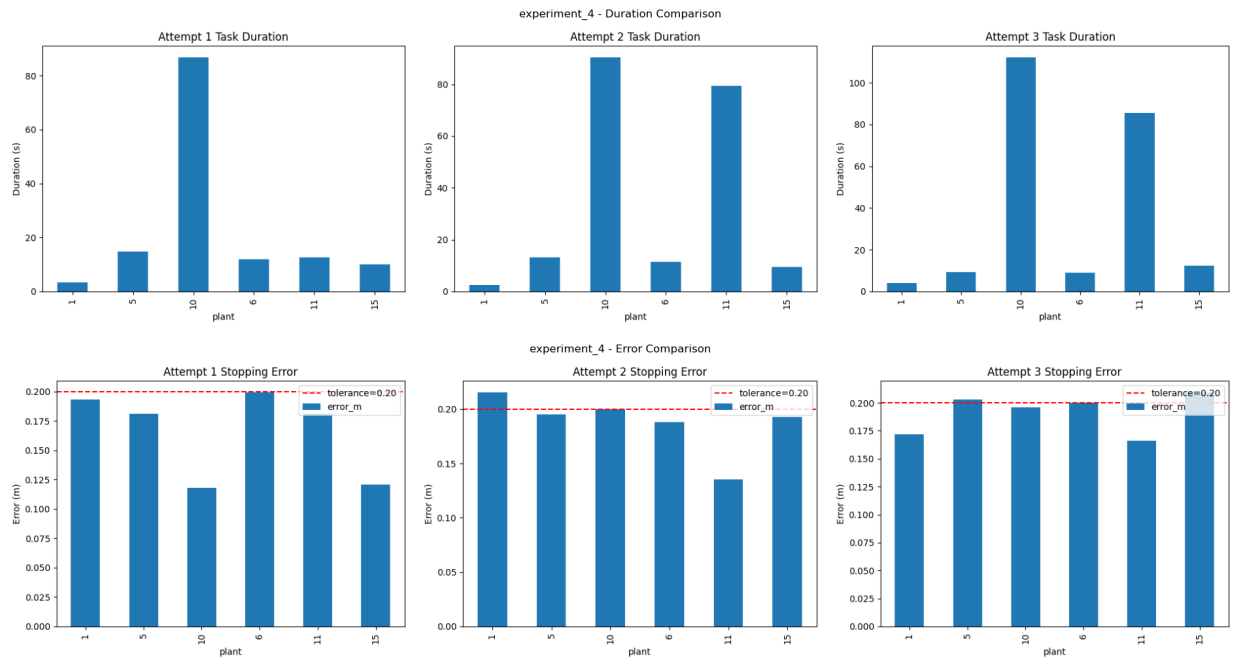


Figure 4: Experiment Type 4: Stop at predetermined plants.

6 Experiment Situations and Notable Events

This section presents selected situations and notable events encountered during the experiments. Each image illustrates a specific scenario, as described in its caption.

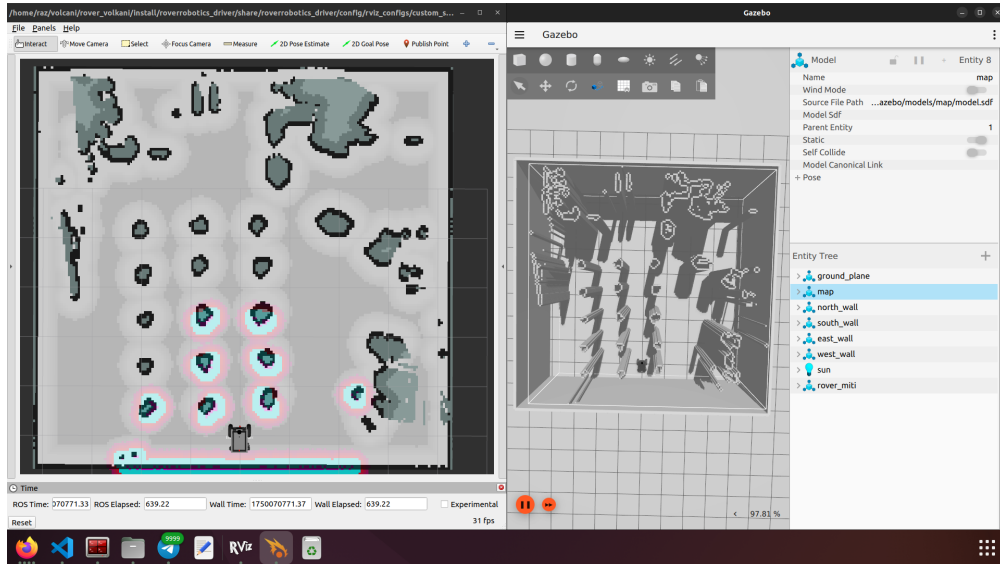


Figure 5: Home position: The robot's initial position in the experiment environment.

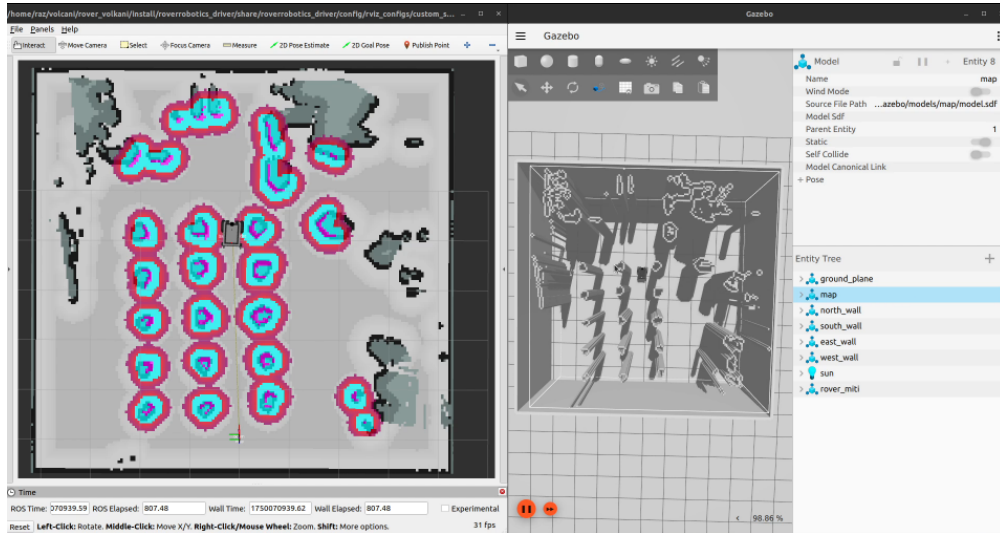


Figure 6: Cost map visualization: The cost map used by the navigation stack, showing obstacles and planned paths.

```

raz@raz-lab: ~$ rosrun rover_volcani
[bt_navigator-10] [INFO] [1750070932.756282526] [bt_navigator]: Goal succeeded
[planner_server-8] [INFO] [1750070933.093159462] [global_costmap.global_costmap]: StaticLayer: Restricting costmap to 163 X 170 at 0.050000 m/pix
[bt_navigator-10] [INFO] [1750070935.157282453] [bt_navigator]: Begin navigating from current location through 2 poses to (4.29, 0.08)
[controller_server-6] [INFO] [1750070935.777876671] [controller_server]: Received a goal, begin computing control effort.
[controller_server-6] [INFO] [1750070935.783108462] [controller_server]: Optimizer reset
[controller_server-6] [INFO] [1750070938.593242487] [controller_server]: Reached the goal!
[bt_navigator-10] [INFO] [1750070938.627638232] [bt_navigator]: Goal succeeded
[bt_navigator-10] [INFO] [1750070941.628674422] [bt_navigator]: Begin navigating from current location through 2 poses to (4.25, 1.24)
[controller_server-6] [INFO] [1750070941.649475460] [controller_server]: Received a goal, begin computing control effort.
[controller_server-6] [INFO] [1750070941.656277275] [controller_server]: Optimizer reset
[controller_server-6] [INFO] [1750070944.683069024] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070947.716410273] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070950.749830896] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070953.783066986] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070956.816309560] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070959.849747511] [controller_server]: Passing new path to controller.
[controller_server-6] [ERROR] [1750070962.883131771] [controller_server]: Failed to make progress
[controller_server-6] [WARN] [1750070962.883205967] [controller_server]: [follow_path] [ActionServer] Aborting handle.
[controller_server-6] [INFO] [1750070962.899216374] [local_costmap.local_costmap]: Received request to clear entirely the local_costmap
[controller_server-6] [INFO] [1750070962.780176896] [controller_server]: Received a goal, begin computing control effort.
[controller_server-6] [INFO] [1750070962.867118801] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070965.900379981] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070968.933697292] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070971.967019466] [controller_server]: Passing new path to controller.
[controller_server-6] [ERROR] [1750070972.867072972] [controller_server]: Failed to make progress
[controller_server-6] [WARN] [1750070972.867162966] [controller_server]: [follow_path] [ActionServer] Aborting handle.
[controller_server-6] [INFO] [1750070972.879065989] [local_costmap.local_costmap]: Received request to clear entirely the local_costmap
[planner_server-8] [INFO] [1750070972.879584660] [global_costmap.global_costmap]: Received request to clear entirely the global_costmap
[planner_server-8] [WARN] [1750070973.101088104] [planner_server]: Planner loop missed its desired rate of 20.0000 Hz. Current loop rate is 4.6296 Hz
[controller_server-6] [INFO] [1750070973.119267879] [controller_server]: Received a goal, begin computing control effort.
[controller_server-6] [INFO] [1750070976.152745282] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070979.186105610] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070982.219450056] [controller_server]: Passing new path to controller.
[controller_server-6] [ERROR] [1750070982.252824291] [controller_server]: Failed to make progress
[controller_server-6] [WARN] [1750070983.252896592] [controller_server]: [follow_path] [ActionServer] Aborting handle.
[controller_server-6] [INFO] [1750070983.269076363] [local_costmap.local_costmap]: Received request to clear entirely the local_costmap
[controller_server-6] [INFO] [1750070983.269539709] [controller_server]: Received a goal, begin computing control effort.
[controller_server-6] [WARN] [1750070983.343231462] [controller_server]: Control loop missed its desired rate of 30.0000Hz
[controller_server-6] [INFO] [1750070985.243289972] [controller_server]: Passing new path to controller.
[controller_server-6] [INFO] [1750070988.276633335] [controller_server]: Passing new path to controller.

```

Figure 7: Experiment 1: The robot performed a prolonged stop at plant 10, as seen in the log. This may indicate a navigation or detection delay at this plant.

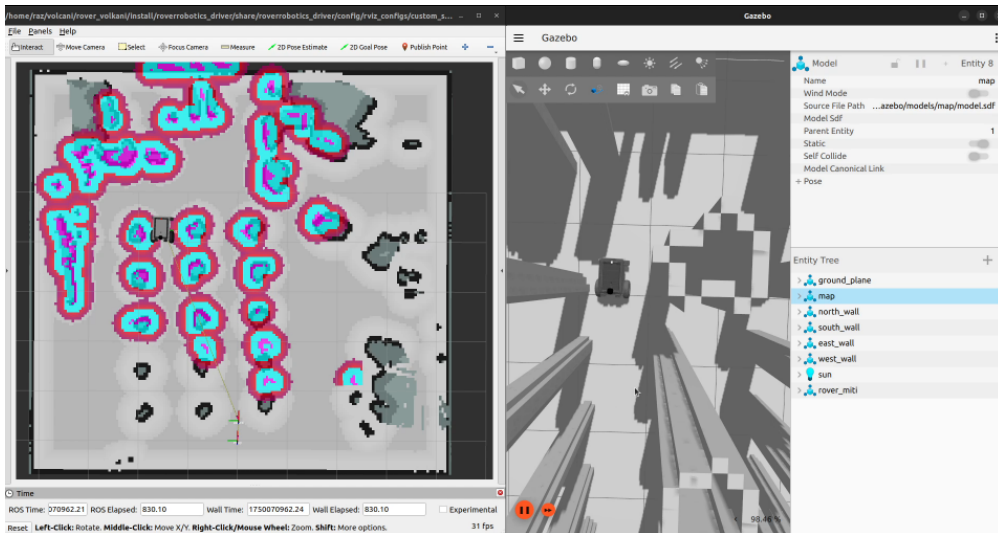


Figure 8: Experiment 1: Simulation view of the robot's long stay at plant 10, confirming the behavior observed in the log.

```

[bridge: [/scan (sensor_msgs/LaserScan) -> /scan (gz.msgs.LaserScan)] (Lazy 0)
[parameter_bridge-3] [INFO] [1750072366.462129250] [ros_gz_bridge]: Creating GZ->ROS
[bridge: [/imu/data (gz.msgs.IMU) -> /imu/data (sensor_msgs/Imu)] (Lazy 0)
[parameter_bridge-3] [INFO] [1750072366.462397828] [ros_gz_bridge]: Creating ROS->GZ
[bridge: [/imu/data (sensor_msgs/Imu) -> /imu/data (gz.msgs.IMU)] (Lazy 0)
[ruby5(which ign gazebo-1) Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use
QT_QPA_PLATFORM=wayland to run on Wayland anyway.
[create-2] [INFO] [1750072366.932828826] [ros_gz_sim]: Waiting messages on topic [/ro
bot_description].
[create-2] [INFO] [1750072367.032677231] [ros_gz_sim]: Requested creation of entity.
[create-2] [INFO] [1750072367.032723754] [ros_gz_sim]: OK creation of entity.
[robo] [create-2]: process has finished cleanly [pid 49022]
[ruby5(which ign gazebo-1) libEGL warning: egl: failed to create dr12 screen
[ruby5(which ign gazebo-1) libEGL warning: egl: failed to create dr12 screen
[ruby5(which ign gazebo-1) libEGL warning: egl: failed to create dr12 screen
[ruby5(which ign gazebo-1) libEGL warning: egl: failed to create dr12 screen
[parameter_bridge-3] [INFO] [1750072550.540966587] [ros_gz_bridge]: Passing message f
rom ROS geometry_msgs/msg/Twist to Gazebo gz.msgs.Twist (showing msg only once per ty
pe)

[planner_server-8] [WARN] [1750073287.408733330] [planner_server]: Planning algorithm
GridBased failed to generate a valid path to (1.48, 1.12)
[planner_server-8] [WARN] [1750073287.408746276] [planner_server]: [compute_path_thro
ugh_poses] [ActionServer] Aborting handle.
[planner_server-8] [INFO] [1750073287.428641979] [global_costmap.global_costmap]: Rec
eived request to clear entirely the global_costmap
[planner_server-8] [WARN] [1750073287.529951830] [planner_server]: GridBased: failed
to create plan with tolerance 0.50.
[planner_server-8] [WARN] [1750073287.530041314] [planner_server]: Planning algorithm
GridBased failed to generate a valid path to (1.48, 1.12)
[planner_server-8] [WARN] [1750073287.530069781] [planner_server]: [compute_path_thro
ugh_poses] [ActionServer] Aborting handle.
[behavior_server-9] [INFO] [1750073287.548999410] [behavior_server]: Running backup
[behavior_server-9] [INFO] [1750073293.749438406] [behavior_server]: backup completed
successfully
[controller_server-6] [INFO] [1750073293.798939406] [controller_server]: Received a g
oal, begin computing control effort.
[controller_server-6] [INFO] [1750073293.809181606] [controller_server]: optimizer re
set

Index -1
[INFO] [1750073119.223898045] [greenhouse_experiment_node]: Final Plant 12: Row -1,
Index -1
[INFO] [1750073119.223907630] [greenhouse_experiment_node]: Final Plant 13: Row -1,
Index -1
[INFO] [1750073119.223916390] [greenhouse_experiment_node]: Final Plant 14: Row -1,
Index -1
[INFO] [1750073119.223925942] [greenhouse_experiment_node]: Final Plant 15: Row -1,
Index -1
[INFO] [1750073119.229133149] [greenhouse_experiment_node]: Waiting for navigate_thro
ugh_poses...
[INFO] [1750073122.851810840] [greenhouse_experiment_node]: Plant 1 -> OK in 3.6s (err
=0.203m [dx=-0.183,dy=0.089])
[INFO] [1750073135.262994177] [greenhouse_experiment_node]: Plant 3 -> OK in 9.4s (err
=0.39m [dx=-0.158,dy=0.117])
[INFO] [1750073144.304696897] [greenhouse_experiment_node]: Plant 5 -> OK in 6.0s (err
=0.20m [dx=-0.182,dy=0.096])
[INFO] [1750073275.567406950] [greenhouse_experiment_node]: Plant 9 -> FAIL in 128.3s
(err=0.941m [dx=-0.751,dy=-0.567])

ording to authority Authority undetectable
Possible reasons are listed at http://wiki.ros.org/tf/Errors%20explained
[INFO] [1750072426.293331723] [rviz2]: Trying to create a map of size 160 x 160 using
1 swatches
[INFO] [1750072426.549936238] [rviz2]: Trying to create a map of size 163 x 170 using
1 swatches
[INFO] [1750072427.441948093] [rviz2]: Trying to create a map of size 163 x 170 using
1 swatches
[INFO] [1750072435.387173006] [rviz2]: Message Filter dropping message: frame 'odori' a
t time 54.219 for reason 'the timestamp on the message is earlier than all the data i
n the transform cache'
[INFO] [1750072435.726953428] [rviz2]: Message Filter dropping message: frame 'odori' a
t time 54.513 for reason 'the timestamp on the message is earlier than all the data i
n the transform cache'
[INFO] [1750073267.443487791] [rviz2]: Trying to create a map of size 166 x 170 using
1 swatches
[INFO] [1750073268.565570086] [rviz2]: Trying to create a map of size 166 x 170 using
1 swatches

```

Figure 9: Experiment 2: Failure at plant 9 due to a collision, as recorded in the log. The robot was unable to complete its approach.

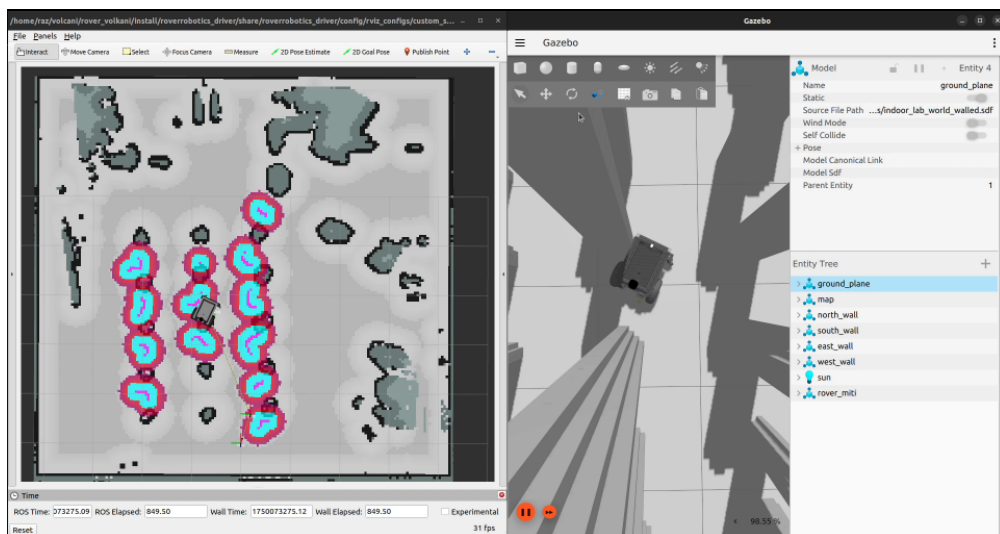


Figure 10: Experiment 2: Simulation snapshot of the collision event at plant 9, showing the robot's position at the time of failure.

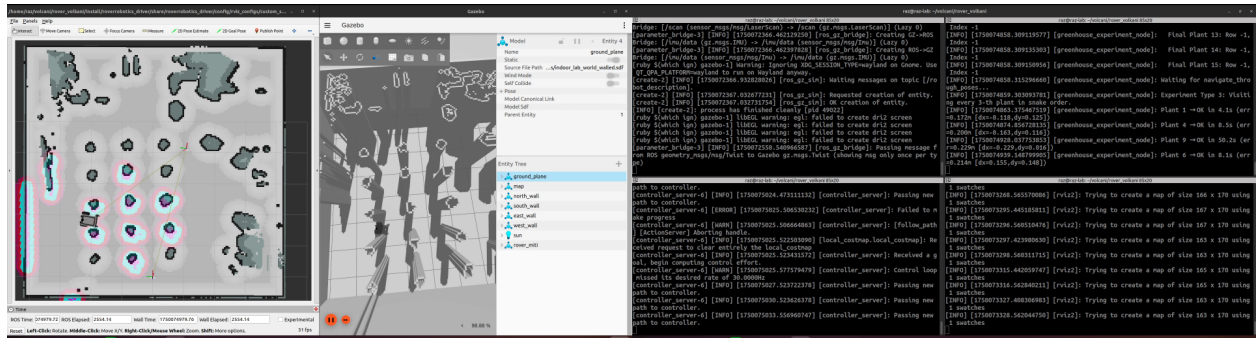


Figure 11: Experiment 3: The robot followed an unwanted path in the middle of row 3, as shown by both simulation and log data. This highlights a navigation anomaly.



Figure 12: Indoor experiment environment: Overview of the Real greenhouse setup used for the experiments.